EE6320 RF Integrated Circuits
Project: LNA Design

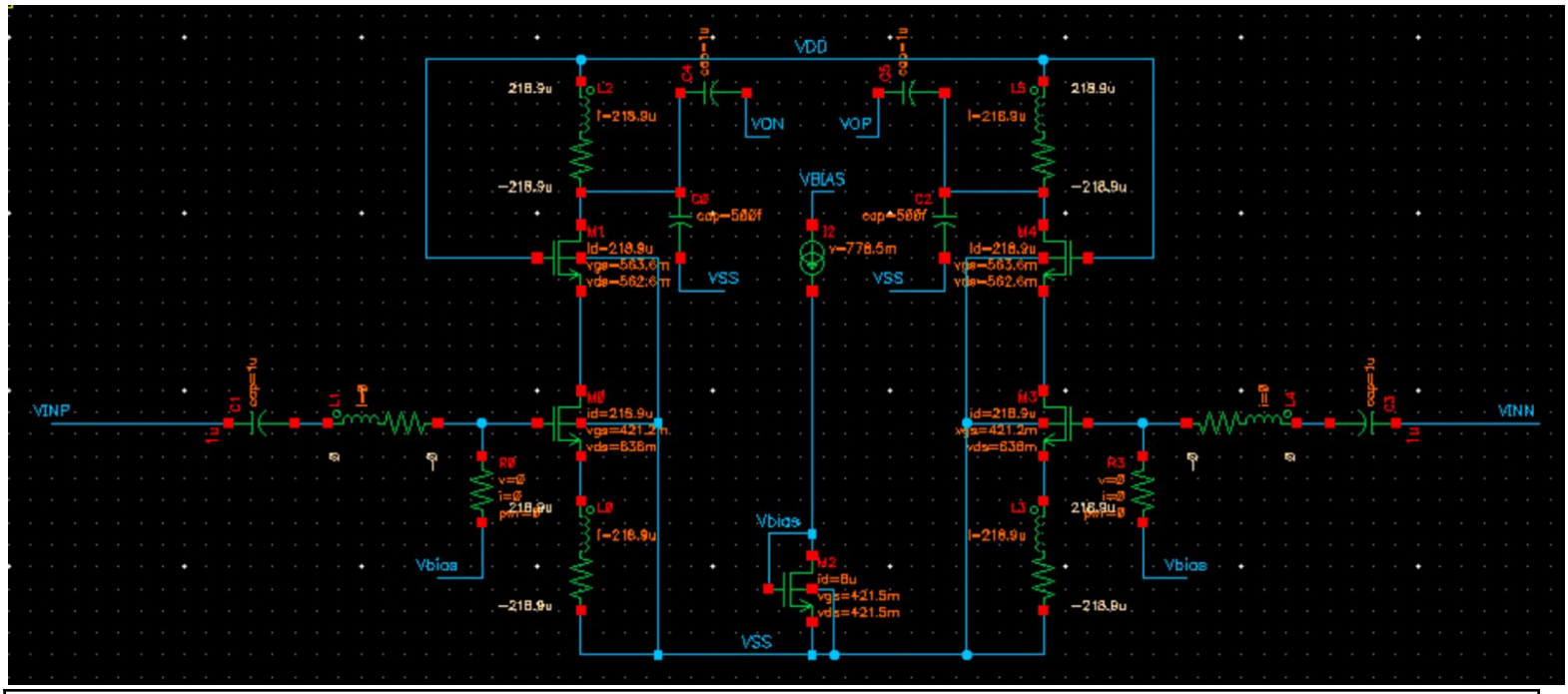
Performance Summary Table

Design metric	Measurement	Simulation Result	Requirement
Input matching	Worst case S11 in the specified band	-11.7292dB	< −10dB
	Band over which S11 ≤ −10dB	3273.9 - 3884.7 MHz	3450 to 3550 MHz
Voltage Gain	Minimum Gain in the specified band	30.551dB	≥ 20dB
	Maximum Gain in the specified band	31.478dB	≥ 20dB
	Gain flatness in specified band [Max-Min Gain]	0.927dB	≤ 1dB
	3dB Bandwidth	226 MHz	-
	Load Capacitance [Differential]	500f F	200f F
Noise Figure	Maximum Noise Figure in the specified band	1.859dB	≤ 2dB
	Minimum Noise Figure in the specified band	1.819dB	-
	Band over which NF ≤ 3dB	2.63 - 4.725 GHz	-
Linearity	IIP3 Tones used	3.5 GHz, 3.501 GHz	-
Power	Input power used for extrapolation	-45 dB	-
	Power of Fundamental Tone at output (at chosen input power)	-13.52 dBm	-
	Power of IM3 Tone at output (at chosen input power)	-83.56 dBm	-
	Extrapolated IIP3	-9.97 dBm	≥ -10 dBm
	LNA DC power consumption [Excluding Bias]	525.36 micro W	Minimize
	Bias circuit power consumption (power supplied by i_bias)	9.6 micro W	Minimize
Other	Sum of all on-chip inductances	10.7 nH	-
	Sum of all off-chip inductances (2 Lg's)	18 nH	-
	Sum of all resistances [Including bias]	200K ohm	-
	Sum of all capacitances [Including AC coupling, excluding load]	4 microF	-
	Simulator Used	ADE L	-

Name: N YASHO SRIKAR ADITHYA REDDY
Roll No: EE23B114

LNA Schematic

<put your schematic picture here from cadence>



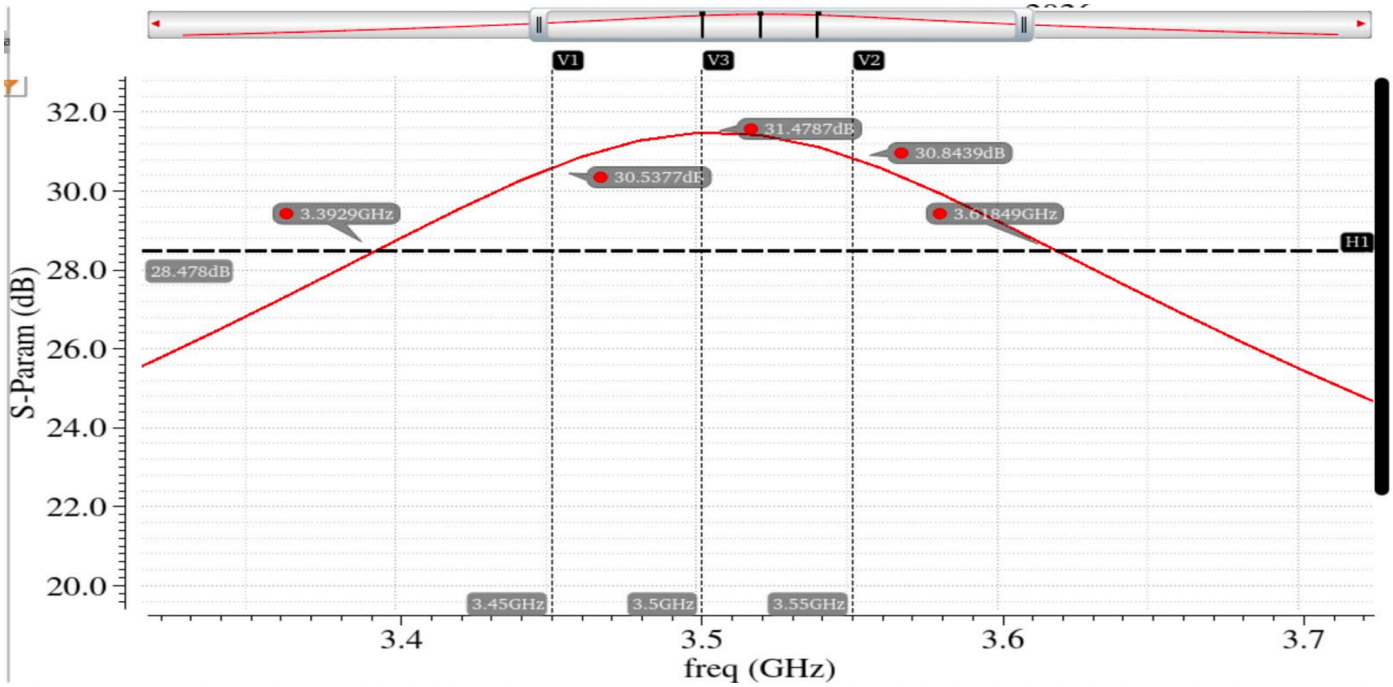
Component Values (one side values)		
Design Component	**Hand Calculated Value**	**Simulated Value**
Lg (gate inductance)	11.58 nH	9 nH
Ls (source inductance)	1.668 nH	1.4 nH
Ld (drain inductance)	4.135 nH	3.95 nH
Rd* (drain resistance)	1818.66 ohm (Q = 20 Ld = 4.135nH)	1563.5 ohm (Q = 18, Ld = 3.95nH)
R_ls* (res. parallel with Ls)	733.6 ohm (Q = 20, Ls = 1.668 nH)	615.7 ohm (Q = 20, Ls = 1.4nH)
Rg* (res. parallel with Lg)	12.732K ohm (Q = 50, Lg = 11.58 nH)	9.89K ohm (Q = 50, Lg = 9nH)

Fixed Constant Parameters

- LNA MOS parameters: W = 125 micro, L = 120 nano
- Current Mirror MOS parameters: W = 6.25 micro, L = 120 nano
- I_bias (current): 8 micro A
- C_coupling: 1 micro F

*** - intrinsic resistance to respective inductors (doesn't come in net resistance on/off the chip)**

<u>Gain Plot</u> (from AC, only for representation)

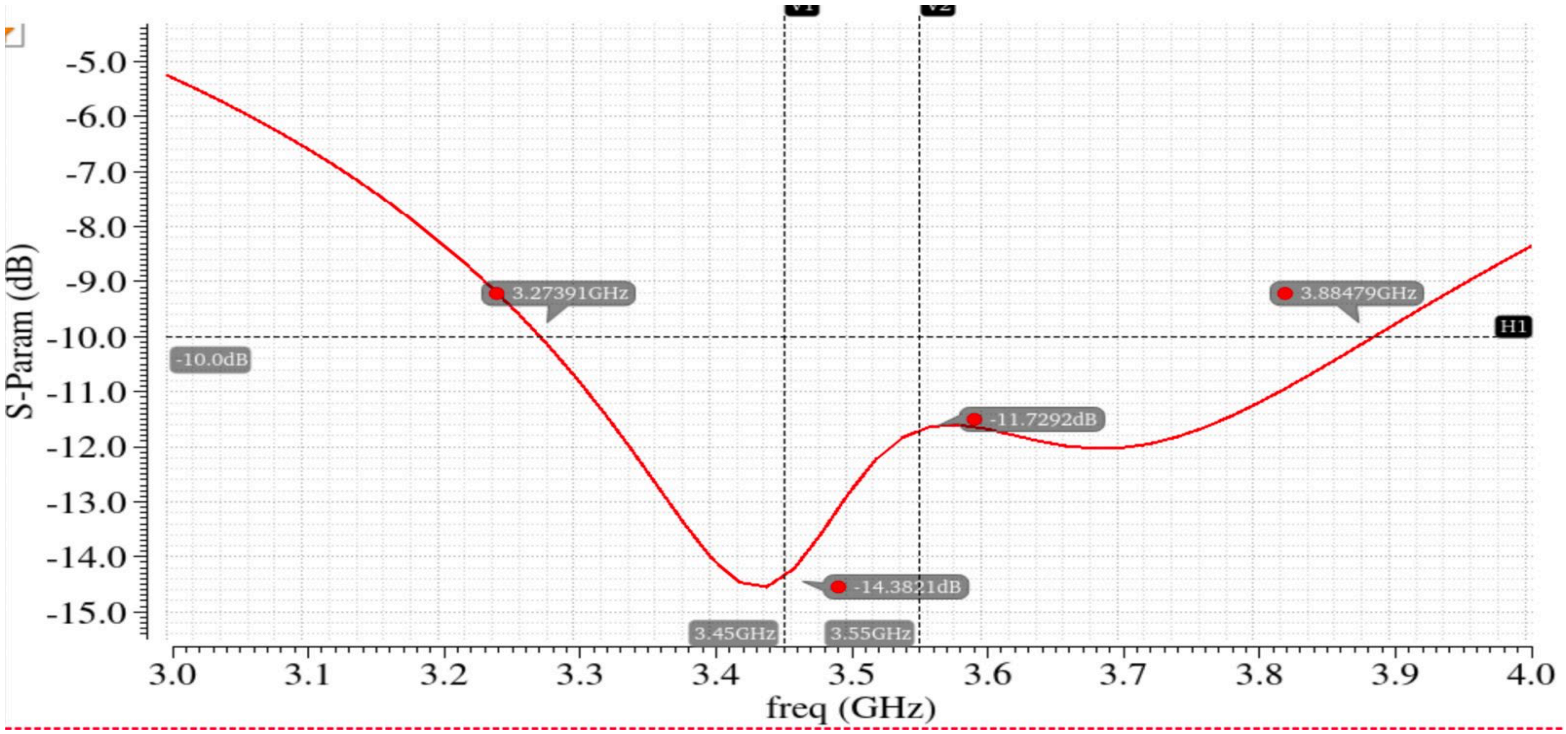


Max Gain through hand calculation: 33.89 dB

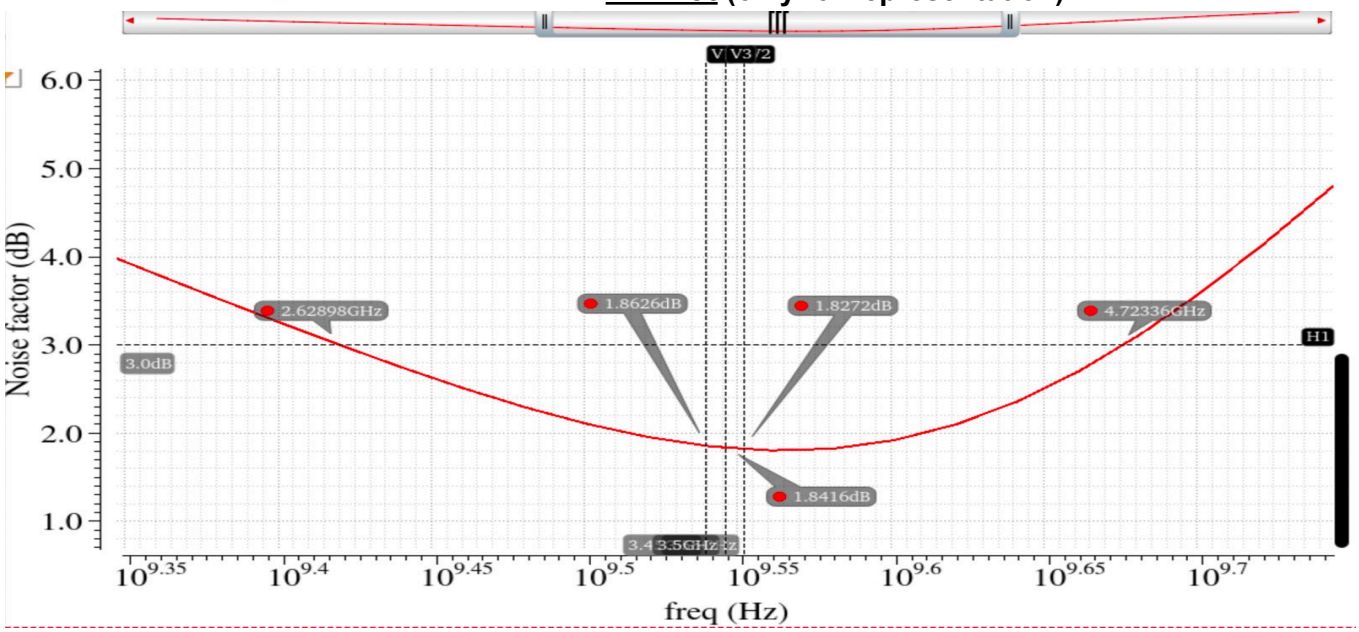
Max Gain from the simulation: 31.378 dB
-3dB Gain comes out to be: 28.378 dB

From the figure, 3dB bandwidth comes out to be: 226 MHz

<u>S11 Plot</u> (only for representation)

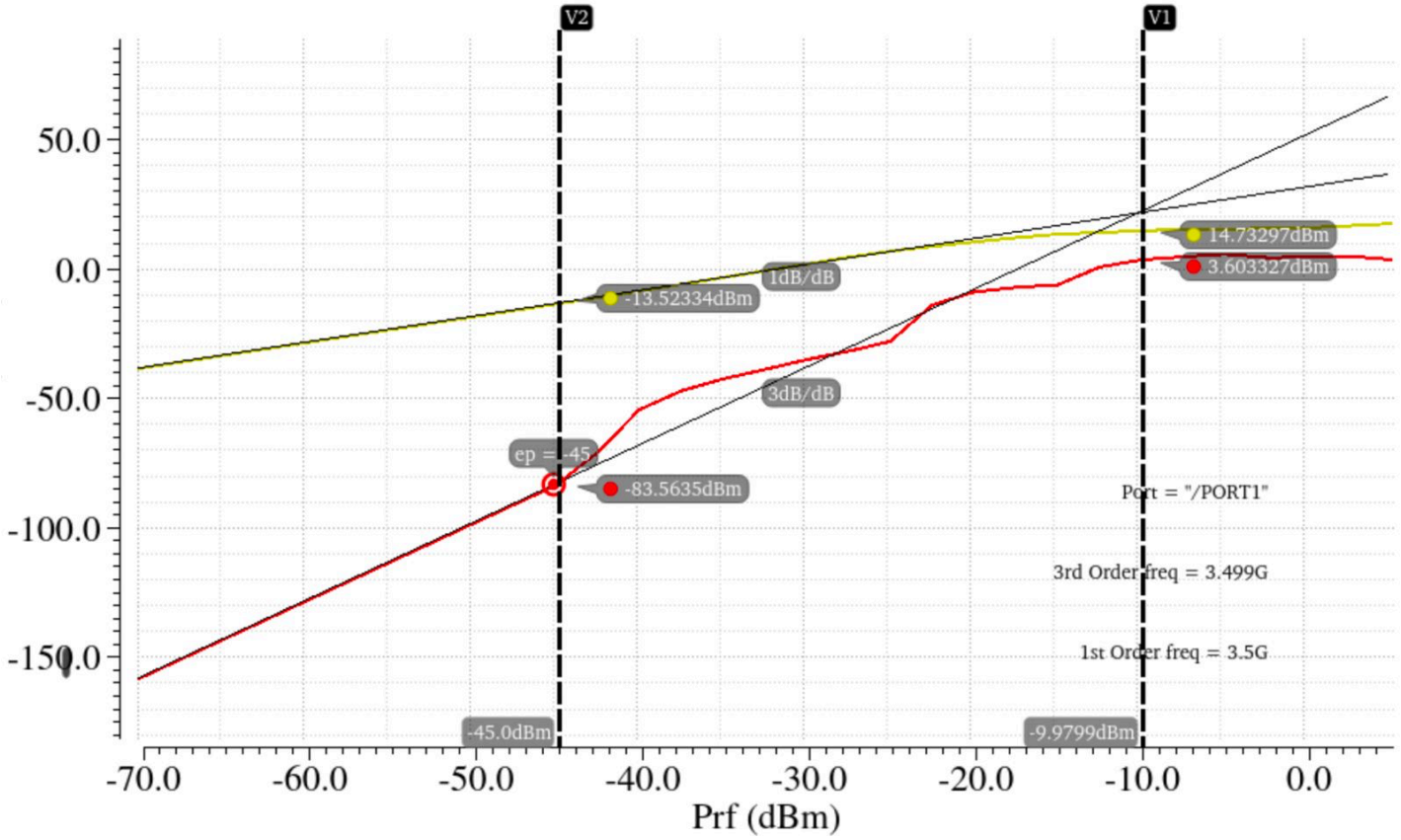


<u>NF Plot</u> (only for representation)



From hand calculations, NF comes out to be: 1.143 dB
From the plot, NF comes out to be: 1.8416 dB

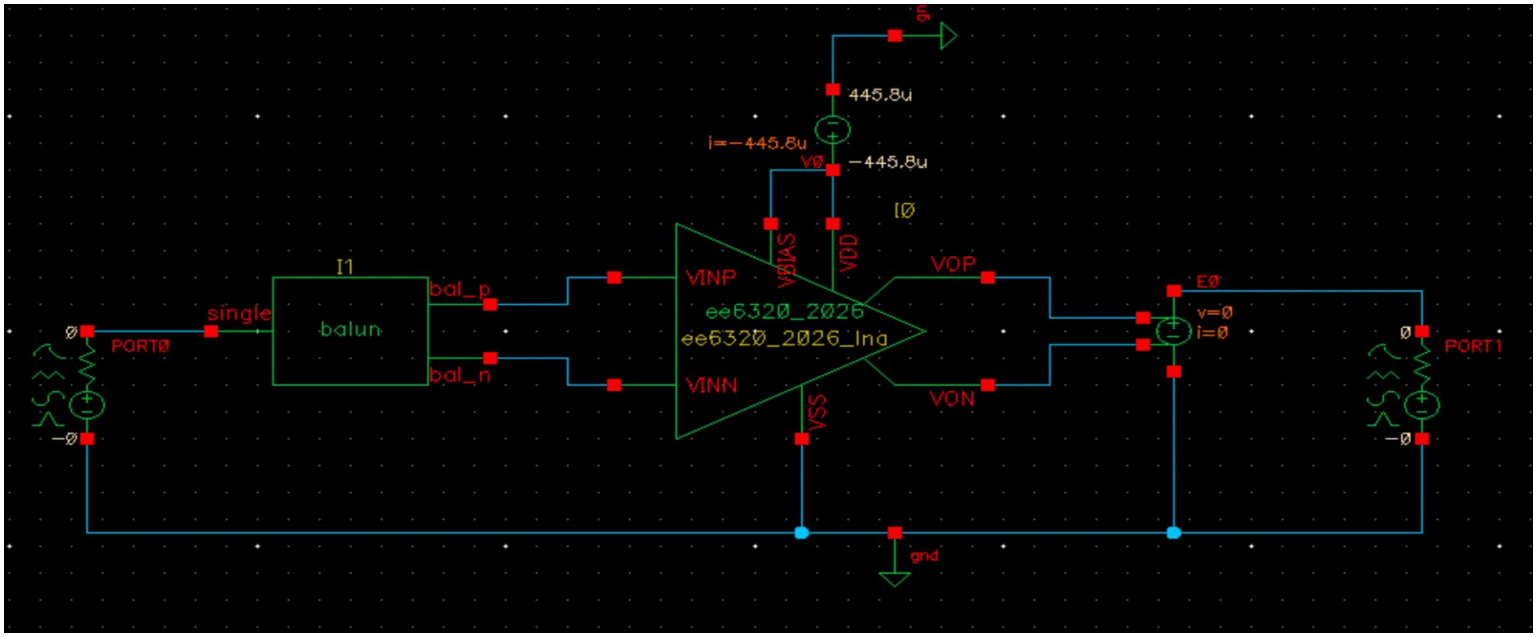
<u>Linearity Plot (IIP3 Computation)</u> (only for representation)
Sweep power from very low powers to very high power and show the extrapolated point



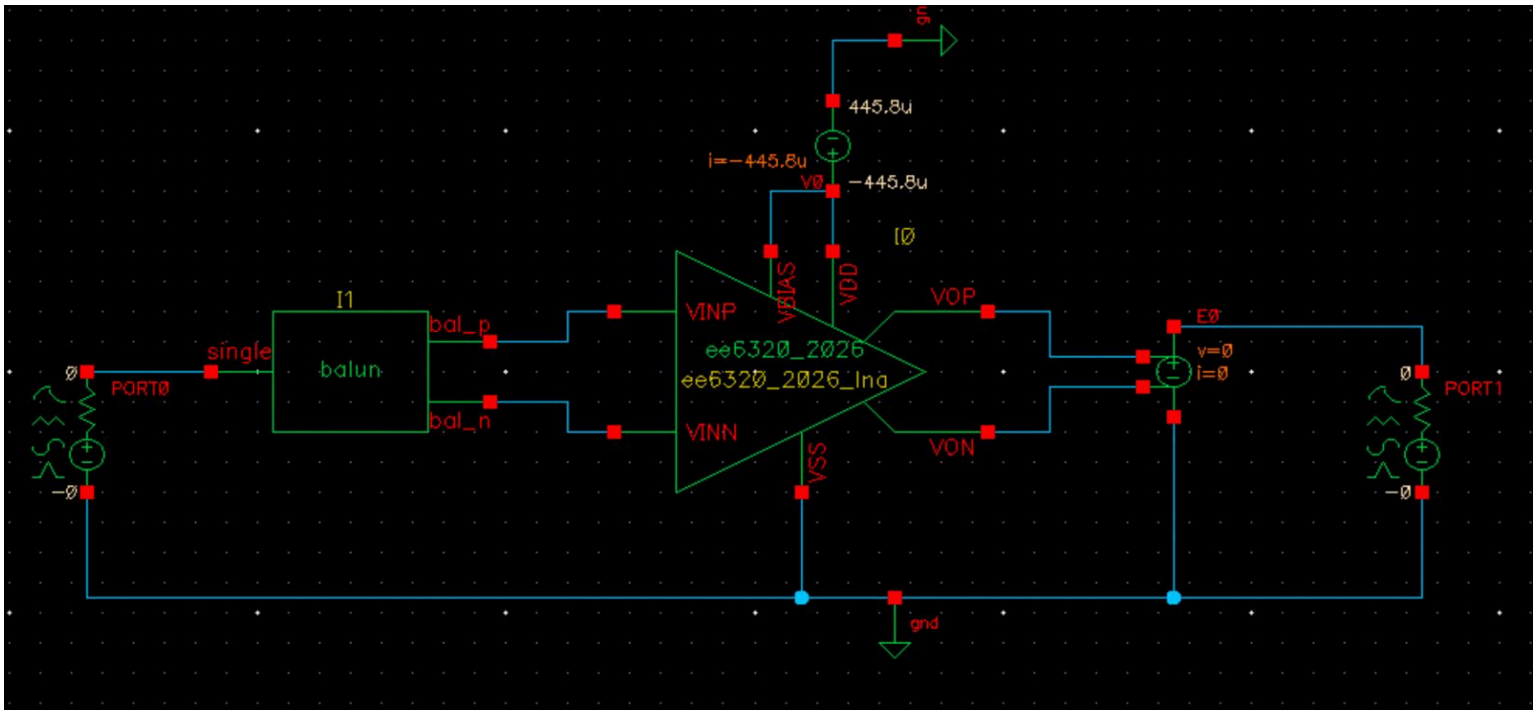
Tones used: 3.5 GHz AND 3.501 GHz
IIP3 point comes out to be: -9.9799 dBm

LNA Testbench

This testbench is used to compute S11, NF, and IIP3 point.



This testbench is used to compute voltage gain.



Comments

Note : R parallel to Inductances are already calculated by taking Q into account. Only the Drain inductor Q is changed for gain flatness. And these resistances are not added into Total on-chip resistance

The gm, cgs values are taken from DC operating values by sweeping though current for a fixed width, not hand calculated.

Tried to prioritize optimizing Power consumption. My 1db compression point is -20.04dB.

Design Procedure

<Brief on the steps followed to arrive at the final design>

Starting with 5u width, I sweeped current vs gm,cgs to choose a high wt for higher noise figure, but IIP3 turned out to be bad. So I kept increasing the width till I have decent head room for IIP3 and NF to tweak other parameters.

Chose 125u width, current 200 uA to calculate the values for other parameters.

After tweaking the parameters I ended up with actual dc current from the differential branch as 218u A.

Used a Width ratio of 20:1 for current mirroring and used a bias current of 8u. Theoretically I should get 160uA but got higher and I tweaked it based on that error.

Hand Calculations done for the Project

For width = 125u, current = 200u

By doing Moschar I got $C_{gs} = 156fF$, $g_m = 4.677mU$

From $\omega_T = g_m / C_{gs} = 29.97 Grad/sec$

Input Impedance $50\Omega = L_s \times \omega_T$

$$\Rightarrow L_s = 1.668nH$$

From resonance of lower LCR tank

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \Rightarrow L = \frac{1}{(2\pi \times f_0)^2 \times C_{gs}}$$

$$L_s + L_g = 13.25nH$$

$$L_g = 11.58nH$$

For load Capacitor $C_L = 500fF$

$$L_d = \frac{1}{(2\pi f_0)^2 C_L} = 4.135nH$$

Assuming Ideal resistance for all inductors,

For Quality factor of '20' for L_s, L_d

$$Q = \frac{R}{\omega L} \Rightarrow R_s = Q \times \omega L_s$$

$$= 2\pi \times f_0 \times 20 \times L_s$$

$$R_s = 733.6\Omega$$

$$R_d = Q \times \omega L_D$$

$$= 20 \times f_0 \times 2\pi \times L_D$$

$$\boxed{R_D = 1818.66\,\Omega}$$

For gate inductor $Q = 50$

$$R_G = 50 \times 2\pi f_0 \times L_G$$

$$\boxed{R_G = 12.732\,k\Omega}$$

$$NF = 1 + \gamma g_m R_s \left[\frac{\omega_0}{\omega_T}\right]^2 + \frac{4R_s}{R_D}\left[\frac{\omega_0}{\omega_T}\right]^2$$

$$= 1 + \frac{2}{3} \times g_m \times 50 \times \left[\frac{\omega_0}{\omega_T}\right]^2 + \frac{4 \times 50}{R_D}\left[\frac{\omega_0}{\omega_T}\right]^2$$

$$\boxed{NF = 1.143\,dB}$$

$$gain = g_m R_D Q_{in}$$

$$= g_m \times R_D \times \frac{1}{\omega_0 C_{gs} \times 2R_s}$$

$$= \omega_T \times \frac{R_D}{2R_s} \times \frac{1}{\omega_0}$$

$$= 49.5$$

$$20\log(49.5) = \boxed{33.89\,dB = Gain}$$

Note

All the images put here (except the test benches) are only for representational purposes, i.e., do not use them as references for your spec values. Use this as a template to make the report (if you are making the report in docs); LaTeX reports are also fine, but maintain the structure as given here.